The Shared Mind

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Volume 12

The Shared Mind. Perspectives on intersubjectivity
Edited by Jordan Zlatev, Timothy P. Racine, Chris Sinha and Esa Itkonen

The Shared Mind

Perspectives on intersubjectivity

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Foreword

Shared minds and the science of fiction

Why theories will differ

Colwyn Trevarthen

It is a pleasure to respond to these essays on the collective story-making of culture: the experience of reality that human beings create together because they are motivated from birth to experiment with the exchange of fantasies and to find meaning in them. Human consciousness has the special gift of imaginative travel through times and places, and it grows through communication of intentions and interests. Language and the practical tools of our society enrich the products of the game, but its causes are in the movements and preferences of embodied minds, minds that have evolved to act in sympathy and to share history and invention, whatever may turn out to be the topic or task. Our common knowledge and perception of ourselves as knowers of meaningful facts depends upon, and grows from, our innate capacity for intersubjectivity.

We know, of course, how involved we are with one another's intentions, thoughts and feelings, and that much of this intimacy in experience cannot be carried in words. The mental life of others is, as Stein Bråten says, "felt immediately" (Bråten 1998). And yet mind science and its ambitious extension in brain science have, and still mainly do, regard us as single heads processing information, storing it up in memory for reprocessing, and transferring it symbolically. Even when we are granted a body that moves, it is a robot that struggles to know other minds by a hopeless effort of "theorizing" or "simulation". Such unsympathetic entities are science fictions. We need a science of the imaginative fictions persons so easily share.

The authors of this book accept that human life and culture is incomprehensible without intersubjective processes – so, the question mark of the title of the editors' introduction in Chapter 1 must be rhetorical, and ironic. It is added, perhaps, because our experimental psychology has inherited and largely still pays homage to a scholastic philosophy of minds as separate experience-registering systems that act, and think, on what they alone perceive. But in the real sociable world every act we make, every feeling, has as much power to move others as it

has to move our self. With compassion we can see causes of actions in another, even causes that they themselves fail to comprehend or control. All teaching and therapy, indeed all cooperative activities, depend on this sympathetic insight into motive impulses and emotions in human moving. I believe that all the inventions of culture, including the evolving languages that distinguish our different societies, and the arts and technologies that are necessary instruments of communal life and treasures of our history, grow from the ability that every young infant has to enter into the co-creation of a proto-conversational narrative with an entranced parent. Our stories of meaning are built on mimetic skills we have inherited from highly sociable animal ancestors, but we are born with new motives for fictional elaboration of rituals. Even our personality, the "who" we are and the narrative of what we have done and known, grants us the role of one protagonist in a social drama where significant others live as supportive allies or contentious rivals (Trevarthen 1993, 1998a, 1998b). Thus we become companions or aliens in relation to a meaningful world. Shared minds create all we know.

Reading this book we sense the authors are glad to be free of a prison built of ideas that are unaware and unsympathetic of how we really live. They present an antithesis to the computational or representational mind, and seek to define the special human mind, which is not just conscious and rational, but has a unique intersubjective awareness that makes up explanations of a shared and artificial world – a mind that builds cultures with power to change nature. Given the exploratory nature of the topic, inevitably, they come to somewhat different conclusions.

Because several authors make generous reference to my research on communication in infancy, and the theory of Innate Intersubjectivity I was rash enough to propose 30 years ago, I feel I should explain the particular scientific experience that supported the project, and the influence of teachers and colleagues who were ahead of me in the story. I was trained as a student of biology to master ways of observing in detail how plants grow and how animals move in adaptive ways. My undergraduate teachers were plant ecologists, physiologists and ethologists. My PhD research was on the experimental neuropsychology of visual consciousness in monkeys with Roger Sperry, who had proposed in 1952 that perceiving must be understood as information picked up to guide moving – that the science of consciousness or mind in the brain should begin by asking how the brain moves the body in intelligent ways (Sperry 1952). My experiments with split-brain monkeys proved that willing to do something can indeed determine what a brain sees.

I began work with infants in 1967, in collaboration in with Jerome Bruner, who wished to examine infant cognition and learning in a different way from Piaget's experiments on infants' object concepts; Berry Brazelton, who was pioneering more sensitive and responsive paediatric care for newborns and their mothers (Brazelton 1979); and Martin Richards, an ethologist of mammalian

maternal behaviour. Our aim was to observe what came about, rather than experiment with *a priori* hypotheses about infant perception or cognition – to record in as complete detail as possible what could be seen and heard when a mother and infant were communicating, and to compare it with what the baby would do when oriented to an inanimate object. We saw complex conversation-like engagements in which both infant and mother exhibited intuitive competence for sharing their impulses, and we realised that there was no science to explain this. At about the same time two other persons – anthropologist and linguist Mary Catherine Bateson, and developmental psychiatrist Daniel Stern – were discovering the same phenomena and attributing them to innate motivations of an intersubjective kind (Bateson 1979; Stern 1971). We were, without knowing one another, exploring out of the psychological box, free to observe the cleverness in infants and their companions and free to speculate about their significance for human relationships and for cultural learning. All of us were entranced by the infants' rhythmic sympathy with a parent's attempts to communicate, and their joint inventiveness.

Through the 1970s, using film and television to record and patiently microanalyse, I charted age-related changes in the play and attributed them to innate motives, the development of new sensory and motor competences in the infant, and sensitive intuitive support from the mother (Trevarthen 1974). I took the term 'intersubjectivity' from an inspiring article Joanna Ryan wrote on the development of 'communicative competence' before language (Ryan 1974), and her comparison of the infant's tactics with those Jürgen Habermas had defined as the intersubjective functions or 'dialogic universals' through which conversational exchanges and cooperative meaning-making are regulated in society (Habermas 1970). At the same time Jerome Bruner led a neo-Vygotskian transformation of developmental and educational theory that gave primacy to collaborative learning in meaningful tasks (Bruner 1968, 1990). Children gain the skills and language of their culture, and learn how to manage the material world in cooperative ways, by way of their will to share purposes, interests and objects (Sinha and Rodríguez this volume).

One of my young colleagues, Penelope Hubley, making a careful longitudinal study of mother-infant companionship in the early 1970s, observed changes from proto-conversations of two-month-olds, through play in games, first of the body, then with shared interest in objects, to the remarkable transformation of the infant's motives at 9 months when the baby became a different kind of partner in intent participation (Hubley and Trevarthen 1979). The enjoyment of playful rituals by six-month olds in games with expressive gestures or toys, enjoying at a new ritualised level the teasing meta-communication that Gregory Bateson had identified as the critical element in animal play (Bateson 1955), was replaced by a more serious intent to do "work" with objects that had some potential for

practical use, which others would acknowledge. Guided by a companion's shifting focus of interest, and by exhortations to complete a little project set by movements of intention, the baby became a self-confident partner – a co-worker. At the same age, about 40 weeks after a full term birth, the baby was a self-possessed and self-conscious performer of many new rituals of social expression. The mutual understanding established in previous months and practiced in games was transformed into what Michael Halliday called proto-linguistic "acts of meaning": vocal and gestural signals of things that might be named (Halliday 1975). We called it Secondary Intersubjectivity. The relevance to cultural learning of this trajectory in growth of the infant's mind was clear, as was the primary importance of the "mutual attention" with a familiar companion. Strangers were too uncomprehending to be trusted in such first steps to a conventional world. And sensitive experimental studies by Lynne Murray proved how important contingent and respectful attention and sympathy of feelings was for the infant to build meaning in another's company (Murray and Trevarthen 1985).

True, as Susswein and Racine (this volume) say, my account of the developments, from Primary Intersubjectivity, through Games of the Person and Games with Objects to Secondary Intersubjectivity, which we continued beyond the first use of words to the Imaginative Play of 2 and 3 year olds, was descriptive. Yes, it was a taxonomy of stages in behaviour, but it was meant to be more than that. It implied and explored a theory, and I sought many kinds of evidence for the "causes" of change, especially causes in the infant's growing mind. I was convinced that the only useful explanation was one that assumed that the fundamental adaptations of body and brain for intersubjectivity were innate, as were the direction and stages of developmental change through the early years, and the learning that was so obviously assisted by the companionship of the parent (Trevarthen 1979, 1989). True to my biological principles, and starting with a theory of how neural systems could generate motives, I looked for explanations in the ontogeny of the brain and body of the child, and for correlations with known age-related changes in brain anatomy and function. It was not difficult to collect evidence that an embryogenetic specification of the motor and sensory functions in somatotopic (body mapping) arrays was essential, as were the theories of experience anticipating motor images of Sperry (1952) and Bernstein (1967). At first we did not have a clue how the intersubjective transfer of these intentional images could be mediated, or what the evident emotional regulations were, but in the last two decades brain science has come some way to closing that gap (cf. Gallese 2005; Pankepp 2005; Barresi and Moore this volume).

One discovery of major significance for any theory of the causal factors or processes of intersubjectivity, whether of humans or animals, is that the rhythmic

timing and modulation of energy in moving is a "code or principle of conduct" that makes motives share-able. A breakthrough in the exploration of human communication before language has come from the demonstration of its special polyrhythmic "musicality" (Trevarthen 1999, 2008). The science of time in the mind or "biochronology" is discovering how impulses regulating the pace and harmony of moving pass from actor to perceiver of action. How, as Ellen Dissnayake claims (Dissnayake 2000), the temporal arts originate in intrinsic dynamic processes, already exquisitely present in a newborn baby, that keep the voluntary and conscious self whole and in well-being, and make them public for sharing in intimacy. Mimesis, which I see as richer already in the neonate than Zlatev (this volume) does (because I am sure that self and other are distinct negotiants in the newborn baby's mind) is the parent of linguistic narrative, as Merlin Donald (2001) proposes, and 'musical semantics' as defined by Ole Kühl sets the stage for reference with symbols (Kühl 2007). I think these are the natural foundations for Itkonen's "normative practices" that keep languages, and other cultural creations, coherent, productive and changing (Itkonen this volume). The story is new and there is plenty of room for different plots, but we have an open prospect and a sense of adventure. The science of the shared mind looks like the best game in town.

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CHAPTER 1

Intersubjectivity

What makes us human?

Jordan Zlatev, Timothy P. Racine, Chris Sinha and Esa Itkonen

1. Introduction

The title of this book, *The Shared Mind*, conforms to a linguistic schema – *"The X Mind"* – that has become common within the interdisciplinary fields of cognitive science and consciousness studies. The current volume thus stands in a line of succession from *The Embodied Mind* (Varela, Thompson and Rosch 1992), *The Discursive Mind* (Harré and Gillet 1994), *The Conscious Mind* (Chalmers 1996), *The Extended Mind* (Clark and Chalmers 1998) and *The Social Mind* (Valsiner and van der Veer 2000). Like some of its predecessors, *The Shared Mind* advances an anti-thesis to the "classical" *Computational* (Jackendoff 1987) or *Representational Mind* (Fodor 1987), with its oft-criticised neglect of the role of the body, phenomenal experience, social interaction and culture.

At the same time, the present volume advances a position (or rather, a set of related positions) that has not been sufficiently explored by its predecessors. Non-human animals also have an "embodied mind", and there are no good reasons to deny that at least birds and mammals also have a "conscious mind" (Edelman 1992). However, although other species may have varying degrees of awareness, they do not seem be fully aware of the subjectivity of others. And whereas human beings go on to engage in discursive practices and rely on material and symbolic culture, both of which have powerful formative effects on the human mind, something more ontogenetically and phylogenetically basic seems required to be able to benefit from these central aspects of human social life. This foundation seems to be provided by a uniquely human capacity for *intersubjectivity*.

In the simplest terms, intersubjectivity is understood by the authors represented in this book as *the sharing of experiential content* (*e.g., feelings, perceptions, thoughts, and linguistic meanings*) *among a plurality of subjects.* Although some non-human species manifest some aspects of the capacity or capacities that make up intersubjectivity, they appear to lack others. On the other hand, no human being is entirely devoid of the human intersubjective potential – even though they

may be delayed or challenged in the expression of some of its manifestations, such as is the case for people with autism.[1] These considerations underlie our bold contentions that the human mind is quintessentially a shared mind and that intersubjectivity is at the heart of what makes us human.

2. Intersubjectivity vs. "Theory of mind"

The hitherto dominant approach in psychology, cognitive science and philosophy has been to analyze what has come to be known as social cognition in terms of a "theory of mind" (or "mentalizing") that purportedly solves the philosophical and developmental problem of "other minds". Consider, for example, the title of a recent volume with an apparently similar theme to the present one: *Other minds: How humans bridge the divide between self and others* (Malle and Hodges 2005). Despite the important empirical findings and hypotheses generated by the Theory of Mind (ToM) approach, it is our contention that its framing of the research question has significantly obscured rather than clarified what needs to be explained. The basic assumptions of the ToM approach can be formulated as follows:

- There is a primary separation between the self and (the minds of) others.
- The individual must bridge this separation either by some form of "theory" or "simulation" of the other's mind, a process that is more or less fallible.
- The main "bodily" structures that are directly relevant for the process are those innate or acquired "modules" engaged in the inferential or simulation processes.
- Cognition develops essentially "from the inside out", with innate or acquired cognitive skills being eventually transferred or projected onto others for the purpose of explaining and predicting their behaviour.

From such a point of departure, it is unsurprising that there appears to be not only a divide, but a veritable gulf between self and others, one that is so wide that it is doubtful whether it could ever truly be bridged. Such a pessimistic assessment of the human condition is hard to justify – how, if it were so, would young children be able to coordinate their basic activities with others, and eventually acquire a shared public language? How could we account for such universal forms of human experience as mutual affection and sympathy? In contrast to the four claims

1. In stating this we are aware that profound and multiple intellectual impairments may raise empirical questions about this claim, but we make it as a generalization with a fundamental theoretical status. We also stress that, even given cases of empirical doubt, our claim does not imply that such individuals should be thought of as not having the status of human beings.

listed above, the contributors to the present volume broadly agree on the following propositions:

- Human beings are primordially connected in their subjectivity, rather than functioning as monads who need to "infer" that others are also endowed with experiences and mentalities that are similar to their own.
- The sharing of experiences is not only, not even primarily, on a cognitive level, but also (and more basically) on the level of affect, perceptual processes and conative (action-oriented) engagements.
- Such sharing and understanding is based on embodied interaction (e.g., empathic perception, imitation, gesture and practical collaboration).
- Crucial cognitive capacities are initially social and interactional and are only later understand in private or representational terms.

The main precursors and originators of these views in the last century were Husserl, Vygotsky and Wittgenstein. Husserl, the founder of phenomenology, has only recently been properly understood in the Anglo-Saxon world to be concerned not with the nature of *private* experience, but with structures of experience which give us a *common life-world*, serving as the pre-condition of any objectivity (Zahavi 2003; Moran 2005). Furthermore, he was the first modern thinker to emphasize the role of the *body* for the emotional tone and the perceptual richness of the life-world, and for our transparent relations with others (cf. Gallagher 2005). For example, he stated:

> I do not first constitute my things and my world solipsistically, then grasp by empathy the other 'I' which too grasps itself solipsistically as constituting its world, and then and only then, the constituted unity of both are to be identified; my self unity (*Sinneinheit*) exists because of the facts that the foreign multiplicity is not different from mine, it is *eo ipso* the same…"
>
> (*Husserliana* 14: 10, translated and quoted by Moran 2005: 225)

Other scholars such as Merleau-Ponty (1962), Scheler (1954) and Schutz (1966) continued this tradition and developed complementary accounts of intersubjectivity (cf. Zahavi 2001) whose common theme is that the basic forms of understanding others are not inferential, but rather *direct* (cf. the chapter by Gallagher and Hutto). Scheler stresses the implications of this for accounts of perception in a way that is reminiscent of J. J. Gibson's (1979) ecological psychology:

> For we certainly believe ourselves to be directly acquainted with another person's joy in his laughter, with his sorrow and pain in tears, with his shame in blushing, with his entreaty in his outstretched hands … And with the tenor of his thoughts in the sound of his words. If anyone tells me that this is not "perception", for it

> cannot be so, in view of the fact that a perception is simply a "complex of physical sensations" … I would beg him to turn aside from such questionable theories and address himself to the phenomenological facts.
>
> (Scheler 1954, cited in Gallagher 2005: 228)

Compare Scheler also to the later Wittgenstein (1980: §570), who similarly attempted to dispel the myth of the isolated subject:

> "We *see* emotion." – As opposed to what? – We do not see facial contortions and *make the inference* that he is feeling joy, grief, boredom. We describe a face immediately as sad, radiant, bored, even when we are unable to give any other description of the features.

Through enigmatic aphorisms such as "Nothing is hidden" and "Understanding is not a mental phenomenon", Wittgenstein highlighted the essential dependence of thinking on *public* criteria, and concentrating on the linguistic aspect of this issue, he rendered the notion of a "private language" self-contradictory (cf. the chapter by Itkonen). Wittgenstein also drew attention to the fact that body, mind and behaviour are different aspects of the unity that we call persons. Thus, although such aspects of persons are non-identical and we therefore cannot reduce one to the other, Wittgenstein argued that they are necessarily and, hence conceptually, related and that we typically talk about one via the other.

Vygotsky was a more multi-faceted thinker, creatively combining philosophy, psychology, literature, primatology and education. From a broadly Marxist perspective (though he was accused of "idealism" by the guardians of Soviet ideological orthodoxy), he famously asserted the general principle of the primacy of social interaction in the development of what he considered to be the specifically human "higher mental functions", such as memory, reasoning and language:

> Every function in the child's cultural development appears twice: first, on the social level, and later, on the individual level; first between people (interpsychological), and then inside the child (intrapsychological). This applies equally to voluntary attention, to logical memory, and to the formation of concepts. All the higher functions originate as actual relations between human individuals.
>
> (Vygotsky 1978: 57)

Like Wittgenstein, the crucial social semiotic mediational "tool" for Vygotsky was language, but he also considered the role of other semiotic resources such as artifacts and gestures (cf. the chapters by Rodríguez and Moro; Sinha and Rodríguez) in the child's cultural development.

3. Perspectives

Although there is a good deal of coherence between the positions of the phenomenologists, Wittgenstein and Vygotsky (as well as those of other classic theorists who feature in the discussions in the following chapters, such as Durkheim, Mead and Bateson) with respect to what they *reject* – that is, the notion of a monadic, individual mind, ultimately incapable of reaching out beyond its confines to the world and others – there are important differences between the positions that they advocate. In a similar vein, while all the authors represented here agree on the crucial role of intersubjectivity in human communication and consciousness of self and other, they offer (as the subtitle of this volume suggests) different answers to questions such as the following:

- What is or are the precise sense or senses of the term "intersubjectivity", at what level of organization does it exist, and how does it relate to other notions of the shared mind such as "common knowledge" (cf. the chapters by Itkonen and Sinha and Rodríguez).
- More specifically, should we understand the term "intersubjectivity" as pertaining primarily to a mental or inter-mental capacity, or to the actual instances of participatory practice that both depend upon, and are instrumental in developing, this human capacity? Are these merely different aspects or emphases, or do they constitute fundamentally different perspectives (cf. the chapter by Susswein & Racine)?
- To what extent is there a species-specific, biological basis for the human capacity for intersubjectivity *per se*, and to what extent is it the consequence of social, ecological and cultural factors (cf. the chapters by Leavens, Hopkins and Bard, Rodríguez and Moro and Sinha and Rodríguez)?
- To what extent is human intersubjectivity brought about by language, and what might be the prerequisite conditions for developing or evolving a capacity for language (cf. the chapters by Gallagher and Hutto, Hutto and Zlatev)?
- Does intersubjectivity involve an irreducibly mental aspect that is accessible to consciousness, or is this an (over-) attribution based on manifest behaviour (cf. the chapters by Brinck and Leavens, Hopkins and Bard)?
- What aspects of human intersubjectivity (e.g., the mutual understanding between two subjects that they are attending to the same object) might play a *causal* role in guiding action and which are definitional rather than causal (cf. the chapter by Susswein and Racine)?

Because they are addressed within the chapters that follow, we will not attempt to answer these questions here. But we wish to highlight the following points of (possible) disagreement between some of the authors, to which the reader may wish to pay special attention:

- The chapters by Gallagher and Hutto, Barresi and Moore and Zlatev adopt stage-based accounts of the development (and evolution) of intersubjectivity, while Brinck argues for more continuous development, involving partially independent capacities.
- The chapters by Pika, Zlatev and Verhagen focus on both continuities and discontinuities between animal and human intersubjectivity, while Leavens, Hopkins and Bard find support for a strong form of continuity, prior to the emergence of language.
- Hobson and Hobson base their account of autism on an impairment of a specifically human capacity for identification with others, while Brinck seeks an account in which more simple skills and developmental patterns combine and interact in order to yield more general cognitive and emotional endowments.
- Susswein and Racine argue that "intersubjectivity" is primarily a taxonomic term, used to group together certain kinds of social interactions which we by definition take to involve one or another form of "experiential sharing", rather than a term denoting hidden mental or neurological processes (in contradistinction to e.g., Barresi and Moore).
- Finally, while most authors adopt a definition of intersubjectivity such as the sharing and understanding of experiential content, Sinha and Rodríguez conclude the volume by stressing the primacy of participatory engagement, and the need to extend "inter-mentality" to encompass "inter-corporeality" and "inter-objectivity".

In another sense of the word 'perspectives', this volume brings together approaches and insights from a variety of disciplines: philosophy, linguistics, primatology, evolutionary theory, neuroscience and typical and atypical human development. The authors do not limit themselves to their disciplinary confines, a fact that strengthens the dialogic aspect of the book. Nevertheless, for the sake of perspicuity, we have organised the contributions thematically into three parts dealing primarily with Development, Evolution and Language respectively.

4. Overview of the chapters

It is appropriate that Part I should focus on ontogenetic development, since it is largely through the path-breaking work of Colwyn Trevarthen (1979) on primary (dyadic) and secondary (triadic) intersubjectivity in the first year of life, and Daniel Stern (1985) on the "interpersonal world of the infant", that the concept of intersubjectivity has emerged as a key issue for the contemporary sciences of the human mind.

Departing from Trevarthen's seminal work, Gallagher and Hutto present an account of the progressive emergence of intersubjective skills in childhood, arguing that understanding others requires neither a "theory theory" nor a "simulation theory" of mind, but is made possible by a sensitivity to bodily movements, gazes, facial expressions and, in secondary intersubjectivity, interactions in pragmatic contexts. In reply to claims that a theory of mind is necessary to account for the more sophisticated interactions of older children and adults involving the concepts of folk psychology, Gallagher and Hutto propose the Narrative Practice Hypothesis, according to which it is through direct encounters with stories about reasons for acting in interactive contexts with caregivers that children become familiar with the core structure of folk psychology. Gallagher and Hutto demonstrate how combining insights from the phenomenological and the Wittgensteinian traditions can yield a productive and novel account of a range of empirical findings, consistent with recent developments in neuroscience.

Barresi and Moore elaborate on this neuroscientific theme, going much further than the now familiar (but rarely explanatory) references to "mirror neurons". They present an up-dated version of their theory of social understanding called Intentional Relations Theory, consisting of four levels, through which first-person and third-person information is "matched", yielding both self-other equivalence and differentiation. Interestingly, the two models of the development of intersubjectivity presented in these two chapters, while formulated independently, appear to be largely compatible. At the same time, the relationship between intentional relations and narrative would merit further consideration, since the latter does not seem to play a crucial role in the model offered by Barresi and Moore. Finally, Barresi and Moore outline an application of their stage or level-based model to autism, suggesting that it is precisely the inability to combine proprioceptive and sensorimotor information about the self with exteroceptive information about others that is at the core of autism spectrum disorders.

Autism is further discussed by Hobson and Hobson, who highlight the pivotal significance of the human propensity to identify with other persons, which they suggest is compromised in children with autism. In reviewing a number of their recent studies, the authors argue that in order to share experiences, typically developing children are psychologically linked to the other person and at the same time differentiated. Hobson and Hobson argue further that these early forms of sharing, and the varieties of communication that they support, provide the foundations for the conceptual understanding of self and other that emerges around the middle of the second year of life. Methodologically, the authors show that our human capacity for reflective intersubjectivity is necessary for analysts to be able to make their judgments in the coding of "overt behaviour", and that such rating is itself a "second-person" phenomenon, rather than a matter of detached observation.

Rodríguez and Moro also adopt the use of a qualitative, "clinical" method in analyzing parent-infant triadic interactions. They draw on Vygotsky and Wittgenstein to show that long before a child is able to produce his or her first symbolic and ostensive uses of objects, the first pointing gestures and the first words, the adult acts *with* the child as a "symbol maker". That is, the adult produces ostensive actions with objects, points to them to make clear his or her intentions, using them in a canonical manner and talking to the child almost constantly. Rodríguez and Moro argue that this implies a long process and a variety of levels of intersubjective adult-baby "agreement" on the use of objects, which serves as a precondition for subsequent social and cognitive development. On a more general level they argue against the assumption (basic to ToM approaches) of the "opacity" of social reality.

Brinck proposes a conceptual clarification of intersubjectivity as "the sharing of experiences", suggesting that it divides into a multitude of sub-concepts depending on the understanding of the terms "sharing" and "experiences". She distinguishes, following Stern (1985), three kinds of shared experiences: emotion, attention, and intention. However, Brinck argues for a more external, behaviour-based way of defining them (thus implicitly disagreeing methodologically with Hobson and Hobson). Brinck argues that different combinations of these forms of intersubjectivity enable corresponding intentionally communicative behaviours, providing a novel explanation of why intentional communication first appears at the end of the first year, despite the fact that its "ingredients" are manifest much earlier. It is, she claims, the ability to decontextualize and combine the various intersubjective skills in a flexible manner that underlies the emergence of intentional communication, rather than this being the culmination of a stage model.

Susswein and Racine take a more deflationary approach than the preceding chapters, exploring the distinction between causal and definitional issues in order

to distinguish between explanations of what an organism is doing and how they are doing it, as well as between different types of causal explanation. They argue that intersubjectivity is a taxonomic rather than a causal explanatory concept, i.e., a technical concept used to classify interactive behaviours and abilities rather than to denote vehicles or causes of those behaviours and abilities. They critically examine the idea that intersubjective engagement involves the sharing of mental states, and argue that the role of mental states and experience in intersubjective engagement is often misconstrued. Finally they apply this approach to human activity to distinguish reflective and practical understanding and consider the meaning of "declarative" pointing in early childhood.

Part II of the volume deals with the capacities for intentional communication and intersubjectivity of non-human primates, and especially of our closest relatives in the animal kingdom, the Great apes; and more generally, with the question of the evolution of human intersubjectivity. Comparative psychology is a field rife with controversies, with persistent disagreement between those who emphasize discontinuities and those who argue for (strong) continuity between non-human and human capacities. This debate is reflected by the diverging positions taken by the first two chapters in this section.

Pika compares the communicative gestures of bonobos, chimpanzees, gorillas and orangutans and shows that Great apes have multifaceted gestural repertoires. She demonstrates that these gestures are performed in multiple contexts and are used flexibly, but unlike those of human children seem to be learned mainly via an individual learning process. Being intentional and referential acts, the gestures of the Great apes display at least a nascent understanding of intentionality, and can plausibly have served as a stepping stone in the evolution of language. On the other hand, Pika also emphasizes the differences between human and ape gestures, suggesting an innate bias for human cultural learning.

In contrast, in analyzing the pointing gesture, Leavens, Hopkins and Bard argue against such an innate bias on the basis of the fact that although both captive and wild apes are sampled from the same gene pool, captive apes spontaneously point without overt training whereas wild apes almost never point. The authors review empirical evidence of the development of pointing in apes and in human children, highlighting the significance of species-typical motor development, culturally-specific patterns of child restraint, and stages of cognitive development for the development what they call "explicit reference". Their major claim is that a capacity for explicit reference emerges in our nearest living relatives when they experience similar circumstances to those of human children, involving the rich emotional contours of human affectivity.

Zlatev distinguishes between five evolutionary (and developmental) levels of intersubjectivity and suggests that the latter has co-evolved with *bodily mimesis*:

the use of the body for communicative and representational purposes. He reviews evidence from primatology to suggest that feral and captive apes are at least to a degree capable of the first two levels (involving e.g., empathy, shared attention and imitation), but not of the third level, "triadic mimesis", which involves an understanding of communicative intentions. Enculturated, language-trained apes, on the other hand, show some aspects of triadic mimesis, suggesting how our predecessors could have bootstrapped themselves to this level without language and may have inherited a biological bias for it. The emergence of language, on the other hand, opened the way to higher levels, allowing (consistently with the proposal of Gallagher and Hutto in Part I) the understanding of "beliefs" and the use of folk psychology.

Hutto develops a similar evolutionary scenario in more detail, arguing that an innate "theory of mind" mechanism is neither necessary nor sufficient to account for the evolutionary course of hominid social interaction. Such a purported "module", he argues, fails to explain the remarkable technical advances and the imitative capacities of Homo ergaster/erectus lying somewhere between those of apes and modern humans. Hutto considers the evidence instead for a Mimetic Ability Hypothesis and invokes the notion of re-enactive imagination in determining what this ancient adaptation was likely to have involved. He argues that this hypothesis provides a better explanation of the kinds of intersubjectivity that would have been necessary for the development of language, thereby undercutting the strongest argument for positing the existence of innate theory of mind modules, namely the support they lend to intention-based semantics.

Part III takes up and develops the theme of the relationship between intersubjectivity and language. It is widely acknowledged that language requires intersubjectivity, most straightforwardly because one needs to know what another is referring to in order to learn a language. But this consensus conceals a good deal of debate concerning the nature of language and its relation to thought and consciousness, and the extent to which language might stretch and reshape our basic intersubjective capacities.

Itkonen opens this section by focusing on the central role of *normativity* for language and linguistics, an issue discussed in brief by Zlatev in Part II, as well as the other contributors of this section. He supports his argument by Wittgenstein's "private language argument", showing that language is impossible without public criteria of correctness. Itkonen further presents and defends the "ontology of the social" as third-level common knowledge, and suggests a dialectical synthesis between "collectivism" and "individualism": common knowledge consist of mental states in particular *configurations*, being therefore both based on and irreducible to (individual) consciousness. Finally, he spells out a number of ramifications of his arguments for theoretical and empirical linguistics and concludes by

pinpointing *a priori* (cf. "devout": Honderich 2006) physicalism ("there is nothing but matter and energy fields") as the roots for the anti-normative bias in linguistics and related fields.

Verhagen argues that intersubjectivity is systemically encoded in natural languages, contrasting this view with the standard "informational" conception of linguistic communication. He reviews constructions at different levels of grammatical organization to show that linguistically coded relations of intersubjective coordination exhibit the specific character of being "rhetorical" or "argumentative". He argues that although these forms may result in cooperation, their immediate function is to influence another person's mind and to make discourse proceed in a particular direction. Because of this, he suggests that language is to some extent analogous with other animal communication systems, which also involve the management and assessment of other organisms, notably conspecifics. On the other hand, he argues for an important discontinuity, in that such management and assessment is oriented in human beings to other minds, rather than to immediate behavior.

Janzen and Shaffer take this theme into a different context, noting that because interlocutors make constant assumptions about what information is active within each other's consciousness, this process creates an interesting challenge in third-party interpretation from one language to another. Their major point is that when an interpreter is introduced into a discourse event this affects the nature of the interchange, since the interpreter will also make assumptions about each of the interlocutors' knowledge states. They use examples from ASL-English interpretation and discuss the notion of "expansions", which have been claimed to be grammatically required in ASL. The authors persuasively argue against this claim. Rather they suggest that understanding *contextualization* as a successful discourse strategy provides a more appropriate approach to the expression of shared and non-shared knowledge and discuss the implications of contextualization for the notion of intersubjectivity in general.

Sinha and Rodríguez end the book where this introductory chapter began, by contrasting the approach to the human mind (and social cognition) based on intersubjectivity with that of "theory of mind", arguing for the advantages of the former. By combining Durkheim's notion of social facts with that of Searle (1995) they argue for the irreducibility of the social to the individual, in a way that converges with the chapter by Itkonen. They also argue against a purely mentalistic conception of intersubjectivity, as a property of the "unmediated mind". Sinha and Rodríguez point out the need to consider inter-corporeality, which extends beyond the body to encompass "inter-objectivity". Objects, they claim, are not only referents of language, but signifiers in their own right, and it is through participatory engagement with the social and material world that children enter the

realm of language. Their chapter concludes by connecting with Gallagher and Hutto's Narrative Practice Hypothesis, stressing the importance of narrativity in the construction of both complex human cognition and shared social cultural identity.

5. Conclusion

As is obvious from these summaries, the "perspectives" expressed in these chapters do not converge on a single, univocal notion of intersubjectivity, but rather point to a complex phenomenon, or a set of related phenomena, in which experiential, behavioural, genetic and neural processes and levels are interwoven in both potentiating and actualizing "what it means to be human". We hope that this introductory chapter has conveyed to the reader our enthusiasm in working on this interdisciplinary and "intersubjective" project. The increasing body of research on social cognition in developmental and comparative psychology, and the prefixing of the term "social" to previously rather individual-oriented fields such as cognitive linguistics and cognitive neuroscience, reflect a changing intellectual context in which we hope that *The Shared Mind* will make a significant contribution to rethinking some of the fundamental questions of our fields. Such a rethinking is an essential, but radically challenging enterprise. The conceptual difficulties encountered by the dominant tradition in the cognitive sciences in attempting to explain the nature of "social cognition", language and communication are not accidental. They stem from the epistemological and methodological individualism inherited from the "possessive individualist" cast of Western culture (and capitalism), and the dominant position accorded in this tradition to natural science and technology *vis-a-vis* the humanities and social sciences (Macintyre 1997; Taylor 1989).

Three of us have collaborated for over 15 years, and our conversations have often revolved around the need for a coherent theoretical statement addressing the centrality of intersubjectivity and normativity for linguistics, psychology and cognitive science. The press of other work, and the difficulty of integrating the various perspectives, has led to the repeated postponement of that venture. In the meantime, however, both the need for, and the timeliness of, a book like the present volume has become increasingly evident.

Tim Racine joined the group at the Jean Piaget Society meeting in Vancouver in the summer of 2005, where the idea for the book emerged from two symposia in which half of the authors represented in the volume participated. Reading (and writing) a large number of drafts of the chapters of this book, and actively

comparing the points of view of the entire volume, helped us, as editors, realize how much our agreement outweighs whatever differences remain. Although this book is a "polyphonic" enterprise, the voices of the different chapters do not always blend harmonically. This is only to be expected in addressing such a quintessentialy interdisciplinary topic as intersubjectivity. In this introductory chapter, we have attempted to show the reader both how a focus on intersubjectivity offers a different (and we believe more productive) perspective to social cognition than the "theory of mind" approach, and to highlight some of the controversies *within* this approach, thereby contributing to defining prospects for further empirical and conceptual investigations. Without meaning to seem unduly naïve, we offer this volume to the reader as a source for reflection on human nature, and on the possibilities for good and ill that are potentiated by the Shared Mind in our shared world.

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PART I

Development

CHAPTER 2

Understanding others through primary interaction and narrative practice

Shaun Gallagher and Daniel D. Hutto

We argue that theory-of-mind (ToM) approaches, such as "theory theory" and "simulation theory", are both problematic and not needed. They account for neither our primary and pervasive way of engaging with others nor the true basis of our folk psychological understanding, even when narrowly construed. Developmental evidence shows that young infants are capable of grasping the purposeful intentions of others through the perception of bodily movements, gestures, facial expressions etc. Trevarthen's notion of primary intersubjectivity can provide a theoretical framework for understanding these capabilities and his notion of secondary intersubjectivity shows the importance of pragmatic contexts for infants starting around one year of age. The recent neuroscience of resonance systems (i.e., mirror neurons, shared representations) also supports this view. These ideas are worked out in the context of an embodied "Interaction Theory" of social cognition. Still, for more sophisticated intersubjective interactions in older children and adults, one might argue that some form of ToM is required. This thought is defused by appeal to narrative competency and the Narrative Practice Hypothesis (or NPH). We propose that repeated encounters with narratives of a distinctive kind is the normal route through which children acquire an understanding of the forms and norms that enable them to make sense of actions in terms of reasons. A potential objection to this hypothesis is that it presupposes ToM abilities. Interaction Theory is deployed once again to answer this by providing an alternative approach to understanding basic narrative competency and its development.

1. Introduction

Our intention in this chapter is to explicate an account of how we come to understand others, without appealing to the dominant theory-of-mind (ToM) approaches of "theory theory" (e.g., Leslie 1987; Gopnik 1993) or "simulation theory" (e.g., Gordon 1986; Goldman 2002). We have elsewhere provided good reasons to

doubt that either of these theories can give an accurate or adequate account of our everyday intersubjective abilities for understanding the intentions and the behaviours of other persons (see Gallagher 2001, 2004, 2006, 2007a, 2007b; Hutto 2004, 2005, 2006a, 2007a, 2007b, 2008). We will briefly summarize that critique here, but our main purpose is to set out a more positive account of just these everyday intersubjective abilities and show that they are not reducible (or inflatable) to the mind-reading or mentalizing described by approaches to social cognition which presume a "theory-of-mind".

This positive account involves three kinds of processes which together are sufficient to deliver the nuanced adult capacity for understanding (as well as for mis-understanding) others. These processes include (1) intersubjective perceptual processes, (2) pragmatically contextualized comprehension, and (3) narrative competence. We argue on the basis of evidence from developmental psychology that the capacity for understanding others is, on average, well established by the time the child reaches four or five years of age, and that it continues to be enriched on the basis of further experience as we become mature adults.

2. A brief critique of the dominant approaches to social cognition

Theory theory (TT) and simulation theory (ST), the standard and dominant approaches to social cognition, share the important supposition that when we attempt to understand the actions of others, we do so by making sense of them in terms of their mental processes to which we have no direct access. That is, we attempt to "mind read" their beliefs, desires, and intentions, and such mind reading or mentalizing is our primary and pervasive way of understanding their behaviour. Furthermore, both TT and ST characterize social cognition as a process of explaining or predicting what another person has done or will do. TT claims that we explain another person's behaviour by appealing to either an innate or acquired "theory" of how people behave in general, a theory that is framed in terms of mental states (e.g., beliefs and desires) causing or motivating behaviour. ST claims that we have no need for a theory like this, because we have a model, namely, our own mind, that we can use to simulate the other person's mental states. We model others' beliefs and desires as if we were in their situation.

Claims that such theory or simulation processes are explicit (conscious) are dubious from a phenomenological point of view. That is, if in fact such processes are primary, pervasive, and explicit, they should show up in our experience – in

the way that we experience others – and they rarely do.[1] The phenomenological critique also rejects the idea, clearly found in TT, that our everyday dealings with others involve an observational, third-person stance toward them – observing them and trying to come up with explanations of their behaviour. Rather, our everyday encounters with others tend to be second-person and interactive.

Claims that the processes described by TT or ST are implicit (or not explicitly conscious) run into a different set of objections. In the case of TT, there is no evidence that such processes are implicit, or even clarity about what precisely that means. Moreover, although TT appeals to false-belief experiments, such experiments are set up to test for explicit rather than implicit theory-of-mind processes (Gallagher 2001) – subjects are asked to explicitly consider the meanings of an observed third-party's behaviour. Implicit approaches to ST appeal to the neuroscience of mirror neurons and shared representations (cf. Barresi and Moore this volume), but there is no justification for calling these subpersonal processes "simulation", since according to ST, simulation involves the instrumental use of a first-person model to form third-person "as if" or "pretend" mental states. In subpersonal processes, (1) there is no first- or third-person (activation of mirror neurons, for example, are considered to be "neutral" in regard to who the agent is) (see e.g., deVignemont 2004; Gallese 2005; Hurley 2005; Jeannerod and Pacherie 2004); (2) nothing (or no one) is *using* a model; and (3) neuronal processes cannot pretend. As vehicles neurons either fire or they don't. More importantly, in terms of relevant content, if they are neutral with respect to first- and third-person, pretence in just these terms (I pretend to be you) is not possible. In effect, simulation, as defined by ST, is a personal-level concept that cannot be legitimately applied to subpersonal processes.[2]

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1. This is not to deny that in some circumstances, for example, in observing puzzling cases of another person's behaviour, we may in fact explicitly appeal to theory or employ simulation. The claim here is simply that most of our everyday interactions are not of this sort. Puzzling cases are the exception.

2. Goldman and Sripada (2005: 208), acknowledging the discrepancy between the ST definition of simulation and the working of subpersonal mirror processes, propose a minimal definition of simulation: "Applied to mindreading, a minimally necessary condition [for simulation] is that the state ascribed to the target is ascribed as a result of the attributor's instantiating, undergoing, or experiencing, that very state. In the case of successful simulation, the experienced state matches that of the target". If this is a necessary condition, it cannot be a sufficient one, because on this minimal definition and without something further, it's not clear what would motivate me to ascribe the state that I was undergoing to someone else. Furthermore, if this were as automatic as mirror neurons firing, then it would seem that we would not be able to attribute a state different from our own to someone else. But we do this all the time. Practically speaking, this proposal also raises puzzles about interacting with more than one other person.

In addition to these phenomenological and logical objections to TT and ST, there is good evidence from developmental psychology that our ability to understand others emerges much earlier than TT or ST would predict. An objection can also be raised against the idea that a general theory (folk psychology) would have sufficient explanatory power to explain the particularities of a large diversity of behaviours found in everyday life, or that it could be very reliable in the face of multiple possibilities for motivation. Similarly it has been objected that running a first-person simulation routine, that is, a process that is based on one's own mental states, seems inadequate to explain the diversity of behaviours found in the world.

These objections throw doubt on TT and ST approaches. The question, however, is whether there is a positive account that can avoid these objections. We turn now to the construction of that alternative account, in three parts: intersubjective perception, pragmatically contextualized comprehension, and narrative competency.

3. Intersubjective perception and interaction

Long before the child reaches the age of four, the capacities for human interaction and intersubjective understanding are already accomplished in certain embodied practices – practices that are emotional, sensory-motor, perceptual, and nonconceptual. These practices include proto-mimesis (Zlatev, this volume), imitation, the parsing of perceived intentions (Baldwin, Baird, Saylor and Clark 2001), emotional interchange (Hobson 2004), and generally the processes that fall under the heading of primary intersubjectivity (Trevarthen 1979). These embodied practices constitute our primary access for understanding others, and they continue to do so even after we attain our more sophisticated abilities in this regard (Gallagher 2001).

In most intersubjective situations, that is, in situations of social interaction, we have a direct perceptual understanding of another person's intentions because their intentions are explicitly expressed in their embodied actions and their expressive behaviors. This understanding does not require us to postulate or infer a belief or a desire hidden away in the other person's mind. What we might

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Is it possible to simulate the neural/mental/emotional states of two other people at the same time if in fact our simulations must be such that we instantiate, undergo, or experience, those two (possibly very different) states? (see Gallagher 2007b). We suggest that these issues would also have to be addressed by Barresi and Moore (this volume) in order to clarify their proposal for a matching system.

reflectively or abstractly call their belief or desire is expressed directly in their actions and behaviours. This phenomenologically direct understanding is likely made possible by the above mentioned complex neuronal processes described as the mirror neuron system(s) and shared representations. In contrast to interpreting these neuronal resonance processes as implicit simulations, which on the functional level would involve cognitive processes over and above the perception of action, Gallagher (2005, in press) has argued that they in fact instantiate a form of *enactive social perception*.

A primary, perceptual sense of others is already implicit in the behaviour of the newborn. In neonate imitation, which depends not only on a contrast, in some sense, between self and non-self, and a proprioceptive sense of one's own body, but also a responsiveness to the fact that the other is of the same sort as oneself (Bermúdez 1996; Gallagher 1996; Gallagher and Meltzoff 1996), infants are able to distinguish between inanimate objects and people. The fact that they imitate only *human* faces (see Legerstee 1991; Johnson 2000; Johnson et al. 1998) suggests that infants are able to parse the surrounding environment into those entities that perform human actions (people) and those that do not (things) (Meltzoff and Brooks 2001). An intermodal tie between a proprioceptive sense of one's body and the face that one sees is already functioning at birth. For the infant, the other person's body presents opportunities for action and expressive behaviour – opportunities that it can pursue through imitation. There is, in this case, a common bodily intentionality that is shared by the perceiving subject and the perceived other. From early infancy humans, and perhaps some animals (see e.g., the studies by Myowa-Yamakoshi 2001; Myowa-Yamakoshi et al. 2004; also cited by Zlatev this volume) have capabilities for primary-intersubjective interaction with others.

The early capabilities that contribute to primary intersubjectivity constitute an immediate, non-mentalizing mode of interaction. Infants, notably without the intervention of theory or simulation, are able to see bodily movement as goal-directed intentional movement, and to perceive other persons as agents. This does not require advanced cognitive abilities; rather, it is a perceptual capacity that is "fast, automatic, irresistible and highly stimulus-driven" (Scholl and Tremoulet 2000: 299). Evidence for this early, non-mentalizing interpretation of the intentional actions of others can be found in numerous studies. Baldwin and colleagues, for example, have shown that infants at 10–11 months are able to parse some kinds of continuous action according to intentional boundaries (Baldwin and Baird 2001; Baldwin et al. 2001). The infant follows the other person's eyes, and perceives various movements of the head, the mouth, the hands, and more general body movements as meaningful, goal-directed movements. Such perceptions give the infant, by the end of the first year of life, a non-conceptual,

action-based understanding of the intentions and dispositions of other persons which does not involve inferences about beliefs or desires understood as mental states (Allison, Puce and McCarthy 2000; Baldwin 1993; Johnson 2000; Johnso, Slaughter and Carey 1998).

Primary intersubjectivity also includes affective coordination between the gestures and expressions of the infant and those of caregivers with whom they interact. Infants "vocalize and gesture in a way that seems 'tuned' [affectively and temporally] to the vocalizations and gestures of the other person" (Gopnik and Meltzoff 1997: 131). Infants at 5 to 7 months detect correspondences between visual and auditory information that specify the expression of emotions (Walker 1982). The perception of emotion in the movement of others, however, does not involve taking a theoretical stance or creating a simulation of some inner state. It is a perceptual experience of embodied comportment (Bertenthal, Proffitt and Cutting 1984; Moore, Hobson and Lee 1997). This kind of perception-based understanding, therefore, is not a form of mind-reading. In seeing the actions and expressive movements of the other person one already sees their meaning; no inference to a hidden set of mental states (beliefs, desires, etc.) is necessary.

The capabilities involved in primary intersubjectivity suggest that before we are in a position to wonder what the other person believes or desires, we already have specific perceptual understanding about what they feel, whether they are attending to us or not, whether their intentions are friendly or not, and so forth. There is, in primary intersubjectivity, a common bodily intentionality that is shared across the perceiving subject and the perceived other. As Gopnik and Meltzoff indicate, "we innately map the visually perceived motions of others onto our own kinesthetic sensations" (1997: 129), and the evidence from recent research on mirror neurons and resonance systems in social neuroscience supports this.[3] Thus, before we are in a position to theorize, simulate, explain or predict mental states in others, we are already in a position to interact with and to understand others in terms of their expressions, gestures, intentions, and emotions, and how they act toward ourselves and others. Furthermore, primary intersubjectivity is not primary simply in developmental terms. Rather it remains primary across all face-to-face intersubjective experiences, and it underpins those developmentally later, and occasional, practices that may involve explaining or predicting mental states in others (see e.g., Stern's (1985) idea of a "layered model" in which

3. In citing Gopnik and Meltzoff's claim about the necessity for innate mappings we are not thereby endorsing their theory-theoretic construal of what this involves. Indeed, much of the evidence developed by Meltzoff and cited by Gopnik and Meltzoff supports the idea of a strong intersubjective perceptual capacity in the infant.

developmentally primary understandings are not "superseded" but remain and operate in parallel to more advanced ones).

4. Pragmatic intersubjectivity

If human faces are especially salient, even for the youngest infants, or if we continue to be capable of perceptually grasping the meaning of the other's expressions and intentional movements, such face-to-face interaction does not exhaust the possibilities of intersubjective understanding. Expressions, intonations, gestures, and movements, along with the bodies that manifest them, do not float freely in the air; we find them in the world, and infants soon start to notice how others interact with the world. When infants begin to tie actions to pragmatic contexts, they enter into what Trevarthen calls 'secondary intersubjectivity'. Around the age of 1 year, infants go beyond the person-to-person immediacy of primary intersubjectivity, and enter into *contexts* of shared attention – shared situations – in which they learn what things mean and what they are for (see Trevarthen and Hubley 1978). Behaviour representative of joint attention begins to develop around 9–14 months (Phillips, Baron-Cohen and Rutter 1992). In such interactions the child looks to the body and the expressive movement of the other to discern the intention of the person or to find the meaning of some object. The child can understand that the other person *wants* food or *intends* to open the door; that the other can *see* him (the child) or is *looking at* the door. This is not taking an intentional stance, i.e., treating the other *as if* they had desires or beliefs hidden away in their minds; rather, the intentionality is perceived in the embodied actions of others.[4] They begin to see that another's movements and expressions often depend on meaningful and pragmatic contexts and are mediated by the surrounding world. Others are not given (and never were given) primarily as objects that we encounter cognitively, or in need of explanation. We perceive them as agents whose actions are framed in pragmatic contexts. It follows that there is not one uniform way in which we relate to others, but that our relations are mediated through the various pragmatic circumstances of our encounters. Indeed, we are caught up in such pragmatic circumstances, and are already existing in reference to others, from the very beginning (consider for example the infant's dependency on others

4. Of course, the fact that another's feelings can be hidden is completely consistent with expressivism of this sort. As Wittgenstein says "One can say He is hiding his feelings. But that means that it is not *a priori* they are always hidden" (Wittgenstein 1992: 35e). The point is that our initial, basic engagements with others are not estranged, even if sophisticated creatures like us are capable of hiding or faking their emotions.

for nourishment), even if it takes some time to sort out which agents provide sustenance, and which ones are engaged in other kinds of activities.

As we noted, children do not simply observe others; they are not passive observers. Rather they interact with others and in doing so they develop further capabilities in the contexts of those interactions. If the capacities of primary intersubjectivity, like the detection of intentions in expressive movement and eye direction, are sufficient to enable the child to recognize dyadic relations between the other and the self, or between the other and the world, something more is added to this in secondary intersubjectivity. As noted, in joint attention, beginning around 9–14 months, the child alternates between monitoring the gaze of the other and what the other is gazing at, checking to verify that they are continuing to look at the same thing. Indeed, the child also learns to point at approximately this same time. Eighteen-month-old children comprehend what another person intends to do with an instrument in a specific context. They are able to re-enact to completion the goal-directed behaviour that someone else fails to complete. Thus, the child, on seeing an adult who tries to manipulate a toy and who appears frustrated about being unable to do so, quite readily picks up the toy and shows the adult how to do it (Meltzoff 1995; Meltzoff and Brooks 2001).

Our understanding of the actions of others occurs on the highest, most appropriate pragmatic level possible. That is, we understand actions at the most relevant pragmatic (intentional, goal-oriented) level, ignoring possible subpersonal or lower-level descriptions, and also ignoring interpretations in terms of beliefs, desires, or hidden mental states. Rather than making an inference to what the other person is intending by starting with bodily movements, and moving thence to the level of mental events, we see actions as meaningful in the context of the physical and intersubjective environment. If, in the vicinity of a loose board, I see you reach for a hammer and nail, I know what your intentions are as much from the hammer, nail, and loose board as from anything that I observe about your bodily expression or postulate in your mind. We interpret the actions of others in terms of their goals and intentions set in contextualized situations, rather than abstractly in terms of either their muscular performance or their beliefs.[5] The environment, the situation, or the pragmatic context is never perceived neutrally (without meaning), either in regard to our own possible actions, or in regard to the actions and possibilities of others. As Gibson's theory of affordances

5. Our understanding of the performance of mimes who work without props depends on their excellent ability to express intentions in their movements, but also on our familiarity with contexts. The mime's talent for expressive movements is clearly demonstrated in contrast to what we often experience in the game of charades or pantomime when we haven't a clue about what the player is trying to represent.

(e.g., Gibson 1979) suggests, we see things in relation to their possible uses, and therefore never as a disembodied observer. Likewise, our perception of the other person, as another agent, is never of an entity existing outside of a situation, but rather of an agent in a pragmatic context that throws light on the intentions (or possible intentions) of that agent.

Theory-of-mind approaches, which involve theory (as an application of folk psychology) or simulation, and which focus on the acquisition of the concept of mental states (like belief) around age 3 or 4 years, miss some basic and important capacities for social cognition. Yet, the acknowledgement of capabilities for understanding others that define primary and secondary intersubjectivity – the embodied, sensory-motor (emotion informed) capabilities that enable us to perceive the intentions of others (from birth onward), and the perceptual and action capabilities that enable us to understand others in the pragmatically contextualized situations of everyday life (from 12–18 months onward) – is not sufficient to address what are clearly new developments around the ages of 2, 3 and 4 years. The "elephant in the room" around the age of 2 years is, of course, language. But if language development itself is something that depends on some of the capabilities of primary and secondary intersubjectivity, language also carries these capabilities forward and puts them into service in much more sophisticated social contexts (on this point, from a different perspective, also see Zlatev this volume).

Do children, upon passing explicit false-belief tests, acquire the final conceptual component needed for their mature understanding of reasons, as is the pervasive claim in the theory-of-mind literature? Or does their newfound understanding of false belief simply equate to a capacity to recognize that the other (whether Maxie, or Sally-Ann, or Snoopy, etc.) has a divergent point of view from their own, and no more? And, what lies at the root of this sort of understanding? Is this sort of mastery of the concept of belief a natural consequence of the maturation of theory-of-mind modules, grounded in introspective acts of ostensive denotation or the product of extensive, evidence-based theorizing on their part? We propose that none of these proposals hold up well under close scrutiny (see Hutto 2008: Chs. 9 and 10). If so it is more plausible to think that an understanding of divergent cognitive perspectives is the result of children beginning to participate in conversations of the kind that require recognition of conflicting points of view. This sort of activity can be seen as a natural extension of those forms of imaginative pretend play that require children to occupy different character roles and adopt personas that are different to their own (Hutto 2008: Ch. 7).

A child's initial understanding of the concept of belief is likely to depend on many things but it is notable that many false-belief tests are presented in the form of a narrative and could be interpreted as tests for a certain level of narrative competency. It also worth observing that the strongest data concerning successful

false-belief performance stems from experiments conducted almost entirely on European and American subjects, whose early lives are awash with folk psychological narratives encountered in fairy tales, children books, comic books, television and films (Richner and Nicolopoulou 2001: 408; Nelson 2003: 22). The form, content and focus of the stories and storytelling practices are much the same in these cultures. Indeed, they even share many of the same canonical 'texts'.

Even more important, we must ask, what role does this mature understanding of false-belief play in the lives of children? And, what drives its development and facilitates its incorporation into larger explanatory schemas of explicitly making sense of actions in terms of reasons (in which attributions of belief plays an important but nevertheless limited part)? In addressing these questions it is vital to be aware, as Carpendale and Lewis (2004: 91) stress, that:

> Proponents of the dominant theories have been notably quiet about what happens in development after the child's fifth birthday. However research that explores whether 5-year-olds can use simple false belief knowledge to make inferences about their own and other's perspectives finds that they singularly fail to do so.

5. Making sense of reasons

The ability and motivation to use one's knowledge of false belief in a wider explanatory context, it seems, is late-developing. It comes into play only after children gain an explicit, practical mastery of the concept of belief. This suggests that false belief understanding is not the crowning moment in their early understanding of other minds; children must develop further still if they are to make sense of actions in terms of reasons. What does this involve?

Let's focus on an example. Someone might ask: Why is Laura going to India? If I don't really know Laura, and if I've never heard her say why she is going to India, then I may attempt to get at her reasons in the third-person. This is surely something we do regularly. This sort of speculative attempt at folk-psychological explanation might run as follows. Laura is a young, American college student. Why do young American college students travel to India? Laura, like many young American college students, may believe that India is a romantic place and that she can learn about Eastern meditation practices there and have an adventure. So Laura might desire to go to India for such reasons. One reaches this conclusion by calling on background knowledge – general knowledge or beliefs about what American college students tend to think and value as well as one's knowledge and beliefs about widely held beliefs about India. The attributed reason may be correct

or incorrect in Laura's case, but lacking detailed information about Laura, one is forced to appeal to generalizations informed by knowledge of an impersonal sort.

Two things are worthy of note. First, this kind of speculation is not likely to be very reliable in most interesting cases. Second, there is no obvious reason to think that the background knowledge or beliefs in question is theoretical. To say that one is operating with theories about India and theories about the belief-forming tendencies of American students in such cases is surely to stretch the notion of theory beyond reasonable limits.

Let's modify the example slightly. If I know Laura, but do not know precisely why she is going to India, I will be able to make a more informed guess about her reasons. Laura is the kind of person who really wants to help children in the third world, so that is *probably* why she is going to India. I will have learnt this about her from my previous exchanges with her or on the basis of what others have told me about her. In this case too, my attribution is knowledge-based but the knowledge in question this time is particular and personal. Although, again, hardly theoretical my attribution remains speculative and suppositional.

Here's a third case. Knowing Laura I may already know her reason for going to India or I might get at it by a much more reliable means. I may know why she is going because she may have already told me so. If not, I could always ask her. Of course, she may be lying or self-deceived, but even acknowledging those possibilities direct conversation is undeniably the most secure route to her reasons.

It is important to stress that in each of these cases the capacity to understand why Laura acted (or might have acted), and our ability to digest these answers is framed by the activity of checking to see if *her reason*, as it were, makes sense. Guessing at or learning of a person's reason is only a small part of the story of our everyday understanding of why others act. It is also necessary to situate and evaluate reasons in wider contexts and against certain normative assumptions. Would it make sense for anyone go to India for that sort of reason? In particular, does it make sense for Laura to go? Is doing so in line with her character, her larger ambitions, her existing projects, or her history? What does it say about her? Does it make her a generous person, an idealist or merely naïve? Understanding reasons for action demands more than simply knowing which beliefs and desires have moved a person to act. To understand intentional action requires contextualizing these, both in terms of cultural norms and the peculiarities of a particular person's history or values.

In this light, reasons for acting are best thought of as "the *elements* of a possible storyline" (Velleman 2000: 28). As such, making explicit a person's narrative is the medium for understanding and evaluating reasons and making sense of